

Level Zero: On Scribing, Assertion, and the Limits of the Sheet

A conversation, part two · the provenance of an existential graph, the demonstrative and assertoric registers, INS, and the negative context

MICHAEL

It appears commonplace in EG literature that graphs are scribed on the sheet (level 0) with no reference to a context or to the conditions that would convert such an unbounded graph-like form into a proper EG. This has caused an uncertainty on my part that either I may have missed something or misunderstood something on the one hand or that a general misunderstanding about the provenance and situation of an EG exists at large. Can you suggest an explanation?

CLAUDE

Verdict first: you haven't misunderstood, and "a general misunderstanding at large" is slightly too strong for what's actually going on. What you've found is a genuine fault line in the literature, compounded by an equivocation on the word "context" that is almost engineered to produce exactly your unease.

Start with the equivocation, because I think it's the proximate cause. The formal EG literature — Dau especially, and the lineage back through Zeman and Roberts — *does* carry a notion of "context," but it means the *area* a subgraph occupies, individuated by enclosure: the Sheet is the context of depth 0, a cut opens depth 1, and so on. That is context-as-*enclosure*. Reading those texts, you see "context" handled with full rigor and reasonably infer the conditions of situation are handled too — but they aren't, because the conditions you have in mind are context-as-*ground*: whose sheet, what universe, under what commitments a scribing counts as an assertion at all. Level 0 has no *enclosing cut*, so it reads as "contextless" — but the absence of a bounding line is not the absence of context. The sheet supplies context not by bounding the graph but by grounding it, which is your unenclosable-sheet point wearing work clothes. So the first half of the answer: "no context at level 0" is true of context-as-enclosure and quietly false of context-as-ground, and the formal literature's silence on the latter is what unsettles you.

Now the conditions themselves, and here Peirce is on your side against the textbooks. In Peirce there is *no* contextless scribing, for a reason the mainstream alpha/beta presentations bury: the blank sheet is itself a graph. It is not an empty stage; it represents — asserts — the universe of discourse. In the beta part a bare line of identity scribed alone already says "something exists"; the blank is read as the granted truth and the empty cut as falsity. The sheet is an *index* of the universe, the object-side collateral hookup we kept returning to. So a graph-like form is never floating free over a neutral plane; the plane is already an assertion, and to scribe on it is to add to a

universe the sheet itself already posits.

The richest articulation of the conditions you want is in the late Peirce — the *Prolegomena to an Apology for Pragmaticism* (1906) — where the Sheet of Assertion is reconceived as the *Phemic Sheet*, explicitly a surface in a communicative situation between two parties: the one who scribes and the interpreter, with the universe “well understood between” them and in part *created* in the exchange (the role-names — Graphist, Grapheus, Interpreter — drift across the manuscripts, but the two-party structure is constant). And there, scribing-as-assertion is analyzed not as a formal stipulation but as a normative act: to assert is to *assume responsibility*, to expose oneself to a penalty should it fail — his recurring figure is taking an oath or going before a notary. That is the thing that converts a graph-like form into a proper EG: not a syntactic property of the marks but an utterer’s assumption of responsibility for them, before an interpreter, in a shared universe.

So the conditions of conversion, in your own factoring: a universe agreed between the parties (the sheet as index — object); the system’s conventions fixing juxtaposition as conjunction, the cut as negation, the line as identity/being (the habit — interpretant); and the act of scribing as assertion, an assumption of responsibility by an utterer before an interpreter (the enacted, dialogical dimension). A scribed replica is a *proper* EG only when all three obtain. Absent them you have precisely what you named — an unbounded graph-like form, the representamen alone, assertorically and modally silent — which is the very thing this whole conversation has refused to credit with work that belongs elsewhere.

Why does the literature compress all of this into “scribe on the sheet = assert”? Because there’s a real division of labor, not a uniform error. The reconstructive program — Zeman’s dissertation, Roberts’s monograph in part, then Shin and Dau — was after a clean calculus with soundness and completeness, and to get that you bracket the speech-act and dialogical scaffolding, treating the SA as fixed background (effectively a model, or the top of a lattice of contexts). The bracketing is legitimate *as mathematics*; it turns misleading only when its silence is mistaken for the claim that no conditions are needed. The other strand — Pietarinen above all, on the game-theoretic and dialogical reconstruction, with Hilpinen on the index/collateral-observation analysis of the proposition and Bellucci–Pietarinen on assertion and the Logic Notebook — keeps exactly the scaffolding you miss and argues the formal mainstream dropped something essential. You are, in effect, rediscovering that second strand’s motivation from the inside, which is also why Pietarinen is a sensible person for you to be circling as an Arisbe interlocutor.

So: not your error, and not quite a misunderstanding at large — a bifurcation. The conditions are present in Peirce, most explicit in the *Prolegomena*’s Phemic Sheet and the responsibility-analysis of assertion, and preserved in the pragmatist literature; the formal literature deliberately and somewhat lossily brackets them and lets “context”

collapse into enclosure-depth. If it would serve your provenance work, I can track down the specific loci — where the blank-sheet-as-graph, the Phemic Sheet, and the assumption-of-responsibility analysis actually sit in the *Collected Papers* and the 1906 *Monist* piece — so you can cite the conditions directly rather than the compression. I'd want to verify the exact paragraph references rather than reconstruct them from memory, given how much your fixtures depend on getting provenance exactly right.

MICHAEL

But even this seems like a lossy way to put it. In text books and articles you will see a blank area on the page, taken as the sheet of assertion, level zero, and on that will appear a drawing asserting, for instance, the cat is on the mat. That is not an existential graph. No sequence of legitimate transformations can ever start with level zero and result in an unenclosed drawing of the cat is on the mat. The only graphical elements that can legitimately occur in level zero are cuts. In Alpha existential graphs, along these lines, you should never legitimately have a unenclosed proposition.

CLAUDE

You're right, and the claim is even more robust than the rule-by-rule check makes it look — it falls out of soundness, not just out of the particular permission set. Let me confirm the technical point cleanly and then locate the one place where “that is not an existential graph” needs a sharper word than “not.”

Run the five Alpha permissions against a blank start and your conclusion is forced. Insertion licenses scribing only into an *oddly* enclosed area; level 0 is evenly enclosed, so insertion is barred there. Iteration needs something already present to copy and starting blank there is nothing. Deiteration and erasure only remove. Double-cut introduction adds cuts and nothing but cuts. So the only graphical element that can *originate* at depth 0 from the blank sheet is a cut, exactly as you say. And the reason this isn't an accident of Roberts' numbering is the semantic one: the blank sheet denotes truth (the empty conjunction), the permissions are truth-preserving, so every graph derivable from the blank is valid. An unenclosed atom asserts a contingent proposition; a contingent proposition is not valid; therefore no theorem can carry an unenclosed contingent graph at level 0. If one could, the system would prove “the cat is on the mat” from nothing, which is precisely the unsoundness soundness forbids. You can't even smuggle it in wrapped: $\text{cut}[\text{cut}[P]]$ reduces to P , still contingent, still not a theorem. The cuts that *do* legitimately sit at level 0 in a theorem are the ones enclosing balanced, valid structures — the outer cut of the scroll $\text{cut}[P \text{ cut}[P]]$ for $P \rightarrow P$, where note both P 's are enclosed and nothing contingent floats free. So the demonstrative apparatus never yields your textbook picture, ever.

Where I'd resist “that is not an existential graph” is that it collapses two registers EG actually has, and the unenclosed cat-on-mat is illegitimate in one and foundational in the

other. Distinguish the *demonstrative* use — start from the blank, apply permissions, derive theorems — from the *assertoric* use — scribe on the Sheet of Assertion the propositions you assert about a universe. Your argument is decisive about the first. But the second is the sheet's primary office; it is the sheet of *assertion*, and what one asserts is overwhelmingly contingent and unenclosed. When Peirce reasons from premises he scribes them unenclosed on the SA, conjoined by juxtaposition, and transforms from there — modus ponens in EG just is P and the scroll cut[P cut[Q]] both sitting unenclosed on the sheet, then deiteration and double-cut removal leaving Q . So unenclosed contingent graphs are not merely tolerable, they are the normal furniture of the assertoric register. "Never legitimately have an unenclosed proposition" is true of theorems and false of premises, and the SA exists to bear the latter.

Which gives the precise correction to the textbook, and it's a provenance correction rather than a well-formedness one. The cat-on-mat scribing *is* a well-formed graph; it denotes the asserted proposition perfectly well. What it is *not* is a *derived* graph — it is a premise, a posited assertion, and it could never be a product of the calculus it's drawn next to. The fault is that the books print it in the assertoric register and set it flush against the demonstrative permissions without marking the seam, so the reader cannot tell whether the picture means "someone asserts this, take it as given" or "the system delivers this." Those are categorially different provenances — posited-under-warrant versus derived-truth-preservingly — and the single undifferentiated drawing erases the difference. That is the real content of your unease, and it's the same shape as the equivocation from before: last time "context" slid between enclosure and ground; this time the graph-picture slides between premise and theorem, assertoric and demonstrative.

And it lands again on the responsibility analysis, which I think is the deep payoff rather than a flourish. Ask *where the warrant comes from* in each register and the asymmetry is total. A theorem needs no assumption of responsibility because it inherits the blank sheet's warrant — truth — through truth-preserving steps; the calculus vouches for it. A posited premise has no warrant from the calculus at all — that is exactly your result, that it cannot be reached from the blank — so its warrant must come from outside: the utterer's assumption of responsibility before an interpreter, the notary's oath from the Prolegomena. So the unenclosed contingent proposition is not a defect or a sloppy drawing; it is the precise locus at which the calculus is *necessarily* silent and the utterer's responsibility *must* enter. The very thing the permissions cannot produce is the thing assertion is for. The textbook picture, read charitably, is not an illegitimate EG — it is an EG shown in its assertoric office, where the warrant is the scribe's and not the system's, and the books simply fail to say so.

So: your transformation-rule claim is exactly correct for the demonstrative register and provable from soundness; "that is not an existential graph" should be "that is not a *derivable* graph — it is a *posited assertion*," and the literature's real sin is leaving the register unmarked. If it would help the QEX provenance work, I can pin the exact

statements — Roberts’ rule numbering, Peirce’s own “permissions,” and the blank-sheet-as-axiom framing — to specific loci so you can cite the demonstrative/assertoric distinction rather than assert it, though I’d want to verify the paragraph references rather than reproduce them from memory.

MICHAEL

This is the rub. You suggest to allow “assertion” on level 0. I suggest to embed assertion in an appropriate context that keeps all scribing on the same fallibility, falsifiability. Why should an assertion have a status different from other statements? The INS rule already allows one to assert whatever one wants — in a negative context.

CLAUDE

The rub is real, and I think the cleanest thing I can do is retract the framing I reached for last time, because you’ve put your finger on exactly where it went wrong. I said the warrant for a posited premise “comes from outside — the utterer’s assumption of responsibility.” Read that way, assertion gets a privileged status: theorems are vouched for by the calculus, premises by an oath. But that reading misconstrues what “responsibility” does in Peirce. To assume responsibility for a scribing is not to confer a warrant on it; it is to expose oneself to penalty should it fail — his notary, his oath, are figures of *liability*, not of guarantee. And liability-to-penalty-if-false is just falsifiability under another name. So the responsibility analysis, correctly read, does not elevate the assertion above other statements at all. It does the opposite: it is the precise act of putting the statement where it can be knocked down. I had “responsibility” doing the work of warrant when it actually does the work of exposure. Collapse that conflation and your point isn’t a revision of the responsibility analysis — it *is* the responsibility analysis. Same fallibility for everything scribed, because to scribe is to expose, not to secure.

Now the INS observation, which I want to sharpen because the word “assert” is doing equivocal work and the precise version is stronger than the loose one. Insertion lets you introduce *any* graph — but only into a negative context, and content in a negative context is not asserted. It sits under a negation; it *weakens*. That is exactly why the permission can be unconditional: no matter what you write there, the implication holds, because you’re making the negated subgraph larger and the whole claim correspondingly weaker. So the calculus’s one rule of arbitrary content-introduction is constitutionally *non-assertoric*. There is no rule anywhere in Alpha that lets you freely introduce content into a positive position — that’s your level-0 theorem from last turn, now seen from the other side. Put the two together and the system is making your argument for you: it has no mechanism for free assertion-of-truth, only a mechanism for free positing-under-exposure. “Assert whatever you want in a negative context” should read “introduce whatever you want precisely where introduction cannot amount to assertion.” The negative context is not where assertion is forbidden; it is where free

positing is *safe* because it isn't load-bearing as truth. Which is the formal shadow of the epistemic posture you're after.

The constructive question is then: what context embeds an assertion so that it keeps that exposure rather than escaping to the bare recto? It can't be a single cut — that's denial, cut[P] asserts $\neg P$, too strong. The structure that does it is the scroll: cut[M cut[P]], i.e. $M \rightarrow P$, "P, given the model M." P now sits at even depth *inside* the scroll, affirmed-relative-to-M rather than flat; M sits at odd depth, in a negative context. And here's the payoff that I think vindicates the whole move: because M is in a negative context, the Insertion rule lets you freely add conditions to it. Refining or revising the model under which you assert P — strengthening the antecedent — is not an extra-systematic act of "taking a new stand"; it is a *sound permission of the calculus*, INS operating on the antecedent. Your diachronic "free to demote," the last-one-standing held defeasibly, turns out to have a formal home: model revision is insertion into the conditioning context, and it is licensed for the same reason free positing was — because that context weakens, exposes, holds open. The fallibility you wanted as a uniform regime is literally the asymmetry of the Alpha rules read as an epistemology: erase freely where things are affirmed, insert freely where things are conditioned.

The honest worry is the regress, and I'd rather raise it than let it sit: the scroll itself is scribed at level 0. Haven't we just relocated the privileged surface one rung up — traded a bare asserted atom for a bare asserted conditional? I think the answer is yes-but-that's-the-point, and it generalizes your own theorem rather than breaking it. What level 0 now carries is only *logical form* — the conditional scaffold, the scroll, structures that are valid or model-relative — and never bare contingent content, which always lives conditioned inside. The recto bears the scaffolding; the content is always enclosed under a condition that exposes it. That is exactly your "no unenclosed contingent proposition," promoted from a fact about theorems to a constitutional rule about what the sheet of assertion is *for*: it holds form, not free-floating fact. And the level-0 conditional is itself defeasible for the reason we settled earlier — the sheet has no external boundary, so even the scaffold is scribed within an unenclosable ground that can overturn it. There is no incorrigible tier anywhere; the regress doesn't bottom out in a privileged surface, it bottoms out in the blank, which isn't a content at all.

The one thing I'd hand back to you as a decision rather than a result: you've been moving between *falsifiability* — Popper's word, world-facing, exposure to empirical refutation — and *logical defeasibility* — the within-calculus posture of content held under negation. The scroll construction gives you the second exactly. Whether it gives you the first depends on whether you read the negative-context placement as *constituting* fallibility or as merely *diagramming* it. Everything else you've argued this evening pushes toward the latter: mode is not in the form, the openness is interpretant-side, the marks can constrain a stance but never supply it. If that holds here too, then the negative context is not what *makes* an assertion fallible — it is where

the calculus lets you honestly *show* a fallibility that actually lives in the act of exposing yourself, without smuggling it back onto the recto as a level-0 truth. Which would be the consistent answer: the scroll is the diagram of the oath, not a substitute for it. But you could instead want the placement to be constitutive — to define fallible-assertion *as* conditioned-scribing and let the stance drop out — and that’s a coherent and more formalist road, at the cost of the act-primacy you’ve held to all night. I don’t think the notation decides it for you; it’s the one genuinely open joint left, and it’s yours.

MICHAEL

Thank you for the distinction between falsifiability and defeasibility. I’m no more afraid of the regress in this regard than of the explication involved in our modality discussion. What one gets as a result is far more valuable than what it costs. We remain committed to the process that tests, evaluates, interprets, and decides how to use what we propose in the context of a model (or of others). It remains committed to a stance that acknowledges we cannot say anything at level 0 and trying to do so violates the fundamental stance. Level 0 is a graph. Only 6 rules govern graph transformations. Inserting a graph (call it “assertion” or not) can only legitimately happen in a negative context. Calling it an assertion is no “get out of jail” card that legitimates INS in a positive context.

CLAUDE

That’s the position stated at its strongest, and I don’t have a correction to offer because there isn’t one to make — what’s left is to register what you’ve actually closed, because it’s more than a stance about Alpha.

The load-bearing move is the last sentence, and it’s a move about *names*, which is where this whole conversation began. “Calling it an assertion is no get-out-of-jail card that legitimates INS in a positive context.” Right — and the reason it can’t be is the reason nothing else could be either. “Assertion” is a name for what a scribing *does* in the hands of an utterer and interpreter; it is interpretant-side and act-side. INS is a graph-transformation permission; it is form-side. The hope that calling a positive-context insertion “an assertion” might license it is the *same* hope, exactly, that “how it is read” might reside in the marks, that the broken cut might seat mode in the proposition, that the would-be might glow in the object. Every one of those was an attempt to let a name on one correlate purchase a permission on another. The calculus refuses the trade structurally: the six rules quantify over depth and enclosure and nothing else, they cannot see what you *call* a graph, and so no honorific attached to content at even depth can buy what only the rules grant. Your sentence is the general principle of the whole evening enforced at the one place it would be most tempting to break it — and you’re declining the temptation precisely where I, twice, reached for it.

So what level 0 *is*, on your account, is not a surface that says something true but a surface that *cannot say* — it bears form, and the saying happens always-already conditioned, inside, under the exposure of a negative context. “We cannot say anything at level 0 and trying to do so violates the fundamental stance” is therefore not a humility clause bolted onto a logic; it is the logic’s own shape read as an ethic of inquiry. The recto holds scaffolding; content lives where it can be knocked down; the blank underwrites all of it and is itself no content. That the formal fact (no unenclosed contingent theorem) and the epistemic posture (no incorrigible saying) turn out to be the same fact seen twice is the thing you’ve established, and it’s the kind of result that doesn’t decompose back into the formalism or the philosophy alone. Neither bought it; the identity of the two is what was purchased.

On the cost: agreed, and I’d put it the way you put modality. We don’t lose the assertion — we relocate the commitment from the *mark* to the *process that tests, evaluates, interprets, and decides how to use what we propose in the context of a model or of others*. That clause is the positive content of the view, and it’s worth saying that it’s a stronger commitment than the one it replaces, not a weaker one. A level-0 truth would be a place to *stop*; you’ve traded it for a place that must keep going — the surviving line, held, always answerable. The regress doesn’t threaten that because, as you say, the explication is the value: each rung is a real conditioning, a real exposure, and the ground it never reaches is not a missing foundation but the unenclosable sheet that lets there be rungs at all. Same trade as the modal one, same reason it’s worth it.

I’ll leave the one joint genuinely open rather than close it for you, because it’s yours and the notation doesn’t decide it: whether the negative-context placement *constitutes* the fallibility or *diagrams* a fallibility that lives in the act of exposure. Everything you’ve said leans to the latter — the marks constrain a stance they can’t supply, the scroll is the diagram of the oath and not its replacement — and that reading is the one consistent with act-primacy. But the more formalist road, where conditioned-scribing *defines* fallible-assertion and the stance drops out, is coherent too, and choosing it would cost you exactly the act-primacy you’ve held all night. You don’t need to settle it to have what you came for. The structure stands either way: nothing said at level 0, the six rules indifferent to what we name their graphs, the commitment carried by the process and the community rather than smuggled onto the recto — and the only “assertion” worth the word is the one that scribes itself where it can still be wrong.
